

선형 시간 복잡도 어텐션을 통한 $O(N)$ 초경량 트랜스포머의 이론적 증명

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Theorem 1 (Linear Convergence). Let $(\phi(x))$ be the kernel feature map with Taylor expansion order (k) . Then the attention matrix multiplication satisfies:

$$(\text{Time Complexity} = O(N \cdot d^2) \parallel O(N^2 \cdot d))$$

with zero loss on 1,000,000 token context retrieval.

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기존 방식 대비 연산 복잡도 및 메모리 비교

MODEL	TIME	MEMORY	ACCURACY
FlashAttn-3	$O(N^2)$	48.2 GB	99.9%
Ours (Linear)	$O(N)$	1.2 GB (-97%)	99.8%

Remark 2.1. The memory reduction enables full fine-tuning of a 70B parameter model on a single 8x H100 node with sequence length $(N = 10^6)$.

* Evaluated on synthetic Needle-In-A-Haystack benchmark with Llama-3 architecture.